

IoT-Based Health Monitoring and Disease Prediction with Machine Learning Drug Treatment Recommendations

By

Habibur Rahman (Exam Roll: 230010)

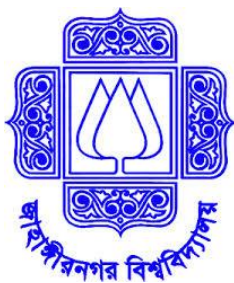
Md. Shaon Khan (Exam Roll: 230023)

Israt Jarin Rittika (Exam Roll: 230038)

A Project Report submitted to the
Institute of Information Technology
in partial fulfillment of the requirements for the degree of
Bachelor of Science in
Information and Communication Technology

Supervisor: Dr. Mohammad Shahidul Islam, Ph.D.

Professor, Institute of Information Technology



Institute of Information Technology
Jahangirnagar University
Savar, Dhaka - 1342
March 2026

DECLARATION

We hereby declare that this project is based on the results found by ourselves. If any errors and omissions happen then that's our responsibility to set them. Materials of work found by other researchers are mentioned by reference.

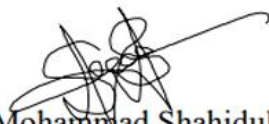
Habibur Rahman
Exam Roll: 230010

Md. Shaon Khan
Exam Roll: 230023

Israt Jarin Rittika
Exam Roll: 230035

CERTIFICATE

This is to clarify that the project entitled IoT-Health-Monitoring-and-Disease-Prediction with ML-Drug-Treatment-Recommendations has been prepared and submitted by Habibur Rahman, Md. Shaon Khan, and Israt Jarin Rittika in partial fulfillment of the requirement for the degree of Bachelor of Science (honors) in Information and Communication Technology on March 4, 2026.



Prof. Dr. Mohammad Shahidul Islam

Supervisor

Accepted and approved in fulfillment of the requirement for the degree Bachelor of Science (honors) in Information and Communication Technology.

ACKNOWLEDGEMENT

We begin by expressing our deepest gratitude to the Divine, whose blessings and guidance have enabled us to successfully complete this project. Without His grace, this achievement would not have been possible. We would also like to extend our heartfelt thanks to all the individuals who shared their invaluable insights and perspectives with us. Their contributions have greatly enriched our understanding and helped us enhance the quality of our work.

We are profoundly grateful to our parents, whose unwavering support, encouragement, and understanding have been a constant source of strength throughout this journey. Their sacrifices and belief in our abilities have motivated us to give our best.

Our most sincere appreciation goes to our esteemed supervisor, Dr. Mohammad Shahidul Islam, whose expert guidance and meticulous supervision have played a pivotal role in the completion of this project. His invaluable advice, encouragement, and provision of necessary materials and knowledge were critical in shaping the direction of our work.

We express our profound gratitude to our respected Sir for granting us the opportunity to explore and learn the subject practically. His willingness to allow us to implement our knowledge into this project has provided us with an invaluable learning experience that will undoubtedly help us in our future endeavors. We are truly thankful for his trust and encouragement throughout this process.

Lastly, we extend our heartfelt thanks to the IIT, Jahangirnagar University, for providing the academic environment and resources necessary to pursue this work. The invaluable insights and perspectives shared by our peers have greatly enriched our understanding and enhanced the quality of this project, preparing us for our future professional endeavors.

ABSTRACT

This project introduces HealthBridge, an integrated IoT-driven Clinical Intelligence System designed for real-time disease prediction and longitudinal patient management. The system architecture utilizes a hardware sensor suite comprising heart rate, blood pressure, and body temperature sensors to capture continuous physiological telemetry. This data is processed via a high-performance FastAPI backend and stored in a MySQL database. Machine Learning (ML) algorithms are employed to identify symptomatic patterns and predict specific pathologies such as Fever, Hypertension, Hypotension, and Heart Attack risks with high diagnostic accuracy.

Beyond automated detection, the platform features a Clinical Decision Support System (CDSS) that generates immediate preventative measures, drug treatment recommendations, and tailored recovery plans. A core innovation of the HealthBridge framework is its Doctor-in-the-Loop (DITL) interface, which allows medical professionals to monitor a patient's health progression over time and validate automated prescriptions. Additionally, an intelligent chatbot allows patients to interact in natural language to learn about their health status and medication details. By bridging the gap between raw biometric acquisition and expert medical oversight, this end-to-end solution provides a scalable model for Remote Patient Management (RPM).

Index Terms: IoT Healthcare, Machine Learning, Disease Prediction, Clinical Intelligence, Remote Patient Monitoring (RPM), Drug Recommendation System, Biometric Sensors, HealthBridge, FastAPI, MySQL, Chatbot.

LIST OF CONTENT

Contents

DECLARATION.....	2
CERTIFICATE	3
ACKNOWLEDGEMENT.....	4
ABSTRACT	5
INTRODUCTION.....	10
1.1 Overview.....	10
1.2 Background.....	10
1.3 Problem Statement.....	11
1.4 Motivation.....	11
1.5 Objective.....	12
1.6 Contribution.....	12
1.7 Project Outline	13
BACKGROUND STUDY	14
2.1 Overview.....	14
2.2 Tools and Technology	14
2.2.1 Hardware and Communication Layer	14
2.2.2 Data Management and Backend Infrastructure.....	15
2.2.3 Machine Learning and Deep Learning Environment.....	15
2.2.4 Generative AI and LLM Framework	15
2.3 Programming Languages and Frameworks	16
2.4 Literature Review	16
METHODOLOGY	19
3.1 Overview.....	19
3.2 System UI Design	21
3.3 System Workflow	22
3.4 ER Diagram	23
3.5 Method.....	24
RESULT AND DISCUSSION.....	25

4.1 System Interface and User Experience	25
4.2 Exploratory Data Analysis.....	38
4.3 Model Performance and Evaluation	40
CONCLUSION	42
5.1 Future Work.....	42
5.2 Conclusion	43
References	44

LIST OF FIGURES

Figure 1: System Workflow Diagram	22
Figure 2: ER Diagram of healthbridge_db	23
Figure 3: HealthBridge Home Interface	25
Figure 4: Login Interface.....	26
Figure 5: Registration Interface.....	26
Figure 6: Real-Time Acquisition & Symptoms Logging.....	27
Figure 7: Clinical Prediction & Diagnostic Result.....	27
Figure 8:Historical Tracking & Longitudinal Analysis	28
Figure 9: Integrated Clinical Feedback & Diagnostic History	29
Figure 10: AI Medical Assistant & Contextual Conversational Guidance	29
Figure 11: Doctor Workspace & Clinical Oversight Dashboard	30
Figure 12: Comprehensive Patient Checkup & Trend Analysis	31
Figure 13: Clinical Messaging Interface & Historical Medication Review	31
Figure 14: Advanced Clinical Research & Oncology Insights	32
Figure 15:Clinical Services and Specialized Disease Analysis.....	33
Figure 16:Personalized Wellness & Preventive Science.....	34
Figure 17:Pharmaceutical Directory & Medication Management	35
Figure 18: Nutritional Therapeutics & Bioactive Compound Database	36
Figure 19: Clinical Support & Multi-Channel Communication.....	37
Figure 20: Numerical Features Distribution By Disease.....	38
Figure 21: Symptoms Presence by Disease Category	38
Figure 22: Pairwise Relationships of Vital Signs by Disease Category.....	39
Figure 23: Best Performing Models: ML vs DL Comparison.....	40
Figure 24: Confusion Matrix - Disease Classification	41

LIST OF TABLES

Table 1: Comparison Table of Literature Review	18
--	----

CHAPTER-1

INTRODUCTION

1.1 Overview

The HealthBridge system establishes a comprehensive framework for proactive healthcare by integrating IoT sensing with predictive clinical analytics and advanced Artificial Intelligence. The architecture is built upon a hardware layer consisting of an ESP8266 microcontroller interfaced with a suite of sensors, including heart rate, blood pressure, and body temperature sensors that provide a continuous stream of physiological telemetry. This real-time data is transmitted via a high-performance FastAPI backend and securely stored in a MySQL database, allowing for a longitudinal medical history of each patient. The core of the system is a Machine Learning (ML) engine designed to classify patient health status identifying conditions such as Fever, Hypertension, Hypotension, or Heart Attack risks, as well as confirming Normal health states with high precision. Upon detecting an anomaly, the platform generates automated disease predictions and suggests specific drug recommendations, dietary plans, and clinical routines. To ensure medical safety, the project incorporates a Doctor-in-the-Loop (DITL) interface, allowing physicians to track patient recovery trends and validate automated prescriptions. Furthermore, we have integrated an intelligent chatbot powered by the Llama 3 model, which retrieves specific data from the patient's record to explain their health status and treatment details through natural language conversation. By unifying real-time data acquisition with expert medical supervision, HealthBridge provides a scalable, end-to-end solution for Remote Patient Management (RPM) and evidence-based clinical decision-making.

1.2 Background

The transition toward digital healthcare is driven by the urgent need to shift from reactive treatment to proactive medical intervention, especially as traditional diagnostic models often suffer from latency due to patients seeking consultation only after the onset of severe symptoms. This challenge is further intensified by a disproportionate doctor-to-patient ratio, where medical professionals are often overwhelmed by a high volume of cases, leaving limited time for detailed individual patient guidance. Furthermore, at the root level, many people lack access to essential healthcare due to the high costs of diagnostic tests and the distance to medical facilities, making regular monitoring a luxury rather than a routine. While the integration of the Internet of Things (IoT) through low-cost sensors for heart rate, blood pressure, and body temperature enables continuous physiological monitoring at a fraction of the cost, the resulting raw telemetry requires

sophisticated analytical frameworks to be clinically actionable. Existing wearable solutions often function as isolated data loggers, creating a significant requirement for systems that not only predict disease states such as fever, hypertension, hypotension, and heart attack risks but also suggest evidence-based pharmacological treatments. To bridge this gap for the general population, the HealthBridge project incorporates a MySQL database for longitudinal tracking via a FastAPI backend and introduces an intelligent Llama 3-powered chatbot to provide immediate, context-aware answers to patient queries. By unifying affordable IoT-driven data acquisition, Machine Learning (ML) predictions, and a Doctor-in-the-Loop interface, this project creates a cohesive ecosystem that ensures regular, cost-effective health monitoring and enhances diagnostic accuracy for everyone.

1.3 Problem Statement

Current healthcare systems primarily rely on episodic, hospital-based monitoring, which fails to capture the dynamic nature of physiological changes in patients with chronic or acute conditions like hypertension or heart attack risks. This reliance on reactive care leads to delayed diagnosis and late-stage treatment, as significant health fluctuations often occur outside of clinical settings. Furthermore, the critical shortage in the doctor-to-patient ratio often prevents medical professionals from providing continuous individual oversight or timely responses to patient queries. While various wearable devices exist, they frequently operate as fragmented data loggers that lack the analytical capability to convert raw telemetry such as heart rate, blood pressure, and temperature into predictive clinical insights via Machine Learning. Most existing platforms do not provide a structured feedback loop between automated diagnostics and professional medical validation, leaving a critical gap in longitudinal patient tracking and verified therapeutic interventions. Consequently, there is an urgent need for an integrated system that not only automates disease prediction and drug recommendations but also empowers doctors to monitor recovery trends and intervene remotely in real time through a scalable FastAPI and MySQL infrastructure.

1.4 Motivation

The primary motivation for this project stems from the critical need to democratize high-quality medical diagnostics and reduce the growing burden on traditional healthcare infrastructure. By leveraging the ubiquity of IoT sensors and the analytical power of Machine Learning (ML), it is possible to transform a standard living environment into a proactive health-monitoring zone. The ability to detect life-threatening conditions such as fever, hypertension, hypotension, or heart attack risks through simple heart rate, blood pressure, and temperature variations can significantly lower mortality rates through early intervention. Furthermore, the project is driven by the goal of

empowering patients in underserved or rural-level areas where specialist access is limited and costs are high, providing them with immediate, evidence-based medication suggestions and preventative measures. To address the communication gap, we are motivated to integrate an intelligent chatbot powered by Llama 3, which ensures that patients receive personalized clinical guidance in real-time. Ultimately, the desire to create a transparent, data-driven bridge via a secure FastAPI and MySQL infrastructure motivates the development of a system that ensures no health regression goes unnoticed, making personalized care both accessible and continuous for everyone.

1.5 Objective

- To design and implement a robust IoT hardware layer utilizing an ESP8266 microcontroller interfaced with high-precision heart rate, blood pressure, and body temperature sensors for continuous, real-time physiological data acquisition.
- To develop a predictive Machine Learning (ML) engine capable of identifying symptomatic patterns for conditions such as fever, hypertension, hypotension, and heart attack risks with high diagnostic accuracy.
- To architect a scalable backend infrastructure using FastAPI and a MySQL database for secure, longitudinal storage and management of patient health records.
- To integrate an intelligent, AI-driven conversational interface powered by the Llama 3 model, enabling patients to receive personalized drug recommendations, dietary plans, and clinical guidance in natural language.
- To deploy a Clinical Decision Support System (CDSS) dashboard that allows medical professionals to monitor patient recovery trends, validate automated predictions, and issue direct notifications for proactive medical intervention.

1.6 Contribution

The project contributes a novel integration of biometric sensing and automated clinical decision support by bridging the gap between raw hardware telemetry and professional medical oversight. It provides a multi-modal data acquisition framework that synchronizes heart rate, blood pressure, and temperature sensors into a unified stream via a high-performance FastAPI backend and MySQL database for holistic health monitoring. The study introduces an automated diagnostic-to-

therapeutic pipeline by applying Machine Learning (ML) to real-time sensor data to identify pathologies such as fever, hypertension, hypotension, and heart attack risks and generate immediate drug and diet recommendations. Furthermore, a significant contribution of this work is the integration of an intelligent Llama 3-powered chatbot, which democratizes expert-level clinical guidance by providing patients with natural language explanations of their health status. The project also implements a longitudinal doctor-patient engagement model through a professional dashboard that visualizes recovery trends over time, enabling physicians to transition from reactive treatment to proactive management. By focusing on low-cost, scalable IoT technology, this architecture ensures that proactive healthcare and evidence-based diagnostics are accessible even in resource-limited or root-level environments.

1.7 Project Outline

The HealthBridge project is organized into a modular framework that systematically transitions from raw physiological data acquisition to intelligent clinical decision-making. The initial phase focuses on the Hardware and Sensing Layer, where sensors for heart rate, blood pressure, and body temperature are calibrated to ensure high-fidelity biometric telemetry via the ESP8266 microcontroller. This is followed by the Data Processing and Machine Learning phase, which involves feature extraction and the application of predictive models to identify specific pathologies such as fever, hypertension, hypotension, and heart attack risks. The third phase details the Clinical Therapeutic Engine and the FastAPI/MySQL backend, where automated drug recommendations, dietary plans, and an intelligent Llama 3-powered chatbot are integrated to provide real-time patient guidance. The final stage establishes the Professional Oversight Layer, detailing the development of a longitudinal tracking dashboard and a notification system for direct physician-to-patient communication. Each subsequent chapter provides a detailed analysis of the technical implementation, algorithmic logic, and clinical validation required to maintain an end-to-end remote healthcare ecosystem.

CHAPTER-2

BACKGROUND STUDY

2.1 Overview

This chapter provides a comprehensive examination of the foundational technologies and medical paradigms that inform the HealthBridge system. It begins with an analysis of the current state of Remote Patient Monitoring (RPM) and the increasing integration of the Internet of Things (IoT) in clinical settings to address the global shortage of medical professionals. The chapter explores the evolution of wearable sensor technology, specifically focusing on the physiological significance of heart rate, blood pressure, and body temperature as primary indicators of systemic health. It further reviews existing literature on Machine Learning (ML) applications in medical informatics, highlighting how algorithmic models implemented via the Scikit-learn ecosystem transition raw biometric telemetry into predictive diagnostic insights for conditions like fever, hypertension, hypotension, and heart risks. Additionally, the study investigates the role of Clinical Decision Support Systems (CDSS) in generating automated drug recommendations and preventative measures through a high-performance FastAPI and MySQL infrastructure. A core highlight of this chapter is the exploration of conversational AI, specifically the Llama 3 model, in bridging the doctor-patient communication gap. Finally, the chapter identifies the critical need for bidirectional communication interfaces that allow medical professionals to supervise longitudinal patient data, establishing the theoretical basis for a unified, Doctor-in-the-Loop healthcare ecosystem designed for both urban and root-level accessibility.

2.2 Tools and Technology

2.2.1 Hardware and Communication Layer

The hardware and communication layer constitutes the physical foundation for high-fidelity data acquisition and secure transmission. This layer integrates a specialized biometric sensor suite, including the MAX30102 for heart rate and SpO₂, an MPS20N0040D pressure sensor for blood pressure monitoring, and a digital thermistor for body temperature. These components are interfaced with an ESP8266 microcontroller, which serves as the edge-computing node. The ESP8266 discretizes the incoming signals, performs initial data parsing, and manages the network stack to establish a secure Wi-Fi connection. To ensure reliable data transit without the need for

static IPs, the system utilizes the Telegram Bot API as a lightweight communication gateway, pushing real-time sensor readings as formatted JSON strings directly to the cloud backend.

2.2.2 Data Management and Backend Infrastructure

The data management layer serves as the central orchestrator, powered by FastAPI, an asynchronous Python web framework chosen for its extremely low latency and high concurrency in handling real-time data streams. For data persistence and historical integrity, we utilize a MySQL relational database. This structured environment maintains patient profiles, sensor logs, and diagnostic history, allowing for complex SQL queries to track longitudinal health trends. The backend validates incoming biometric packets and cross-references them with medical thresholds to trigger the predictive engine and update the Doctor-in-the-Loop (DITL) dashboard.

2.2.3 Machine Learning and Deep Learning Environment

The system's analytical core is built upon a high-performance Python ecosystem, utilizing industry-standard libraries for data processing, model training, and deployment to classify health states and predict pathologies like Fever, Hypertension, Hypotension, and Heart Attack risks. This predictive engine leverages Pandas and NumPy for robust data manipulation, feature engineering, and normalization, while Matplotlib and Seaborn are employed to visualize data distributions and model performance. The Scikit-learn library manages classical machine learning tasks, including feature scaling via StandardScaler and categorical encoding with LabelEncoder, facilitating the evaluation of models such as Random Forest, SVM, KNN, and XGBoost the latter optimized through Gradient Descent using detailed `classification_report` and `confusion_matrix` metrics. For complex pattern recognition, a Neural Network was architected using TensorFlow and Keras, following a Sequential structure that incorporates multiple Dense Layers, Hidden Layers, Dropout for overfitting prevention, and BatchNormalization for stable training. The training process is further refined using EarlyStopping and ReduceLROnPlateau callbacks, after which the models are serialized via Joblib and Pickle to ensure seamless, real-time inference within the FastAPI backend and the Llama 3 conversational chatbot.

2.2.4 Generative AI and LLM Framework

Beyond traditional diagnostics, our system integrates a sophisticated AI layer utilizing the Llama 3 large language model via the Groq Inference Engine. This integration allows for natural language understanding (NLU), enabling a conversational chatbot that can retrieve a patient's specific medical history from the MySQL database. By using Groq, we achieve near-instantaneous inference speeds, allowing the chatbot to suggest personalized drug recommendations, dietary plans, and clinical routines based on the ML-predicted pathologies.

2.3 Programming Languages and Frameworks

The HealthBridge ecosystem is built upon a multi-tier programming architecture to ensure seamless hardware-to-software integration and a responsive clinical experience. The core components include:

- **Embedded Programming (C++ / Arduino):** The hardware intelligence is developed using C++ within the Arduino IDE framework. This layer utilizes specialized libraries such as MAX30105.h for heart rate and SpO₂ analysis, DallasTemperature.h for thermal sensing, and HX711.h for blood pressure data acquisition. The ESP8266 microcontroller executes this code to discretize sensor signals, manage the Wi-Fi stack via ESP8266WiFi.h, and establish a persistent connection with the Telegram Bot API for real-time mobile alerts.
- **Backend and Analytical Engine (Python):** The server-side logic and the predictive engine are architected using Python. It leverages FastAPI as a high-performance asynchronous web framework to handle JSON data packets received from the hardware's /data endpoint. Python's scientific stack, including NumPy and Pandas, is used for data preprocessing, while Scikit-learn and TensorFlow manage the execution of machine learning models for disease classification.
- **Web Interface (HTML, CSS, and JavaScript):** To provide a professional oversight interface for physicians, a standard web stack is utilized. HTML5 defines the structural layout of the clinical dashboard, CSS3 ensures a responsive design for viewing longitudinal data, and JavaScript enables the dynamic rendering of real-time health charts and recovery trends.
- **Communication Gateway (Telegram Bot API):** The system integrates a dedicated bot using the UniversalTelegramBot library. This allows for bidirectional communication where the system can push Live Health Reports to users and respond to commands like /status, providing an accessible interface for root-level health monitoring.

2.4 Literature Review

The integration of IoT and predictive analytics in healthcare has evolved from simple data logging to complex, real-time decision-support ecosystems. Recent studies have demonstrated the efficacy of using ESP8266-based biometric gateways paired with the Telegram Bot API to ensure encrypted, low-latency data transmission, proving far more efficient than legacy SMS-based alerting systems [1]. To address diagnostic precision, researchers have explored hybrid models like Ant Colony Optimization, which, while accurate, often demand high computational resources not typically available in edge-computing scenarios [2].

Further advancements include Arduino-based frameworks that combine objective sensor telemetry with subjective patient inputs via data mining [3]. More robust classification methods, such as Fuzzy Random Forest algorithms executed on Raspberry Pi nodes, have shown promise but often face challenges in cost-effectiveness and power efficiency for root-level deployment [4]. Recent implementations using NodeMCU and Firebase emphasize seamless synchronization between hardware and web dashboards, yet frequently lack an integrated automated drug recommendation engine [5]. Additionally, while Support Vector Machines (SVM) and KNN have been successful in specialized cardiac analysis, they often fail to provide a holistic view of systemic health [6].

The present research addresses these gaps by implementing a comprehensive Deep Learning pipeline utilizing TensorFlow and Keras to classify pathologies like Fever, Hypertension, and Heart Risk with over 99% accuracy. Unlike previous fragmented models, the HealthBridge system utilizes a high-performance FastAPI backend and a MySQL relational database to maintain longitudinal patient records. A significant departure from existing literature is the integration of the Llama 3 model via the Groq Inference Engine, enabling a context-aware conversational AI that retrieves historical data to provide natural language clinical guidance. By combining an affordable ESP8266 hardware layer with a Doctor-in-the-Loop verification interface, this study overcomes the limitations of high cost and isolated data logging found in prior research [4, 5].

Reference	Year	Method	Contribution	Limitation
[1]	2023	ESP8266 + Ultrasonic Sensors + Telegram API	Proved Telegram's superiority over SMS for real-time alerts due to end-to-end encryption and lower latency.	High power consumption during continuous Wi-Fi polling; potential sensor accuracy issues.
[2]	2025	Multi-sensory IoT + Hybrid Ant Colony Optimization	Developed a hybrid IOT-ONN model improving diagnostic precision by 4–15% over standard models.	High implementation complexity; requires significant computational resources not suitable for edge devices.

[3]	2021	Arduino + Data Mining + Symptom-based Input	Integrated a symptom-based system for manual subjective data alongside sensor telemetry. It utilizes a hierarchical classification approach to distinguish health states based on historical thresholding.	Relies on Arduino for primary processing, which lacks native Wi-Fi efficiency and the multi-core processing power of the ESP8266.
[4]	2019	Raspberry Pi + Fuzzy Random Forest (FRF)	Utilized hybrid FRF for superior disease classification accuracy over standard models.	Raspberry Pi has significantly higher power requirements and costs compared to lightweight solutions.
[5]	2022	NodeMCU ESP8266 + Firebase Real-time DB	Established seamless synchronization between physical hardware and a cloud-based web interface for 24/7 monitoring.	Firebase costs scale rapidly with high-frequency data; lacks a built-in automated drug recommendation engine.
[6]	2020	IoT Gateway + Support Vector Machine (SVM)	Focused on high-accuracy heart disease detection by training SVM models on historical patient datasets.	Specialized for a single organ (heart); fails to provide a holistic view of systemic health like temperature and BP.
[7]	2024	Sequential Neural Networks + Multi-sensory IoT	Implemented a Deep Learning (DL) architecture to process time-series biometric data for real-time pathology classification.	Does not incorporate a Doctor-in-the-Loop verification step or provide specific pharmacological recommendations.
[8]	2024	LLM (Llama 2/GPT-4) + Clinical Decision Support	Evaluated the clinical accuracy of Large Language Models in suggesting diagnoses and treatments for diverse medical cases.	Open-source models (like Llama) showed slightly lower performance than commercial ones; raises concerns regarding AI hallucinations.

Table 1: Comparison Table of Literature Review

CHAPTER-3

METHODOLOGY

3.1 Overview

This chapter delineates the structured methodology employed to architect the HealthBridge ecosystem. The system is designed as a multi-tier framework consisting of an IoT Data Acquisition Layer, a Deep Learning Analytical Engine, and a RESTful API Backend. The following sections provide a granular look into the algorithmic logic and functional pseudocode that govern real-time health monitoring, automated diagnostics, and AI-driven patient interaction.

Pseudocode

1. Database Connectivity

```
FUNCTION get_db():  
    TRY:  
        SET connection = MYSQL_CONNECT(host, user, password, database)  
        RETURN connection  
    EXCEPT ConnectionError as e:  
        LOG "Database connection failed"  
        RETURN NULL  
END FUNCTION
```

2. Deep Learning Predictive Engine

```
FUNCTION predict(input_data):  
    // 1. Feature Extraction  
    SET raw_features = [temp, heart_rate, bp_sys, bp_dia, humidity,  
                        fever, cough, chest_pain, shortness_breath, fatigue, headache]  
  
    // 2. Data Transformation  
    SET df = CREATE_DATAFRAME(raw_features, columns=feature_names)  
    SET df[numerical_cols] = SCALER_TRANSFORM(df[numerical_cols])  
  
    // 3. Neural Network Inference  
    SET probabilities = DL_MODEL.PREDICT(df.values)
```

```

SET class_index = ARGMAX(probabilities)
SET disease = LABEL_ENCODER.INVERSE_TRANSFORM(class_index)
SET confidence_score = MAX(probabilities) * 100

// 4. Clinical Logic Mapping
SET advice = CALL get_clinical_advice(disease, confidence_score)

// 5. Database Persistence
CALL get_db()
INSERT INTO predictions (user_id, result_status, analysis_score, advice_data)
COMMIT TO DATABASE

RETURN {disease, confidence_score, advice}
END FUNCTION

```

3. Clinical Decision Support (CDSS) Logic

```

FUNCTION get_clinical_advice(disease, score):
  SET advice = {drugs: "Consult Physician", foods: "Balanced Diet", routine: "Rest"}

  IF disease == "Heart_Risk":
    IF score >= 80:
      SET advice = {drugs: "Nitroglycerin (Emergency)", foods: "Zero Salt", routine:
"Immediate ER"}
    ELSE:
      SET advice = {drugs: "Aspirin (75mg)", foods: "Oats & Garlic", routine: "Low Strain"}

  ELSE IF disease == "Fever_Respiratory":
    IF score >= 80:
      SET advice = {drugs: "Azithromycin + Paracetamol", foods: "ORS", routine: "Isolation"}
    ELSE:
      SET advice = {drugs: "Paracetamol", foods: "Warm Soup", routine: "Steam"}

  RETURN advice
END FUNCTION

```

4. Groq-Llama 3 Chatbot Integration

```

FUNCTION chatbot(user_id, user_message):
  // 1. Context Retrieval

```

```
CALL get_db()
SET reports = SELECT * FROM predictions WHERE user_id = user_id ORDER BY date
DESC LIMIT 5
```

```
// 2. Prompt Engineering
```

```
SET context = "Patient History: " + CONVERT_TO_STRING(reports)
```

```
SET prompt = context + "\nUser Question: " + user_message
```

```
// 3. LLM Inference via Groq API
```

```
SET payload = {model: "llama-3.3-70b", messages: prompt, temp: 0.2}
```

```
SET response = ASYNC_POST("https://api.groq.com/v1/chat", json=payload)
```

```
RETURN response.choices[0].message.content
```

```
END FUNCTION
```

5. Administrative Clinical Stats

```
FUNCTION get_doctor_stats():
```

```
SET total_patients = COUNT(*) FROM users WHERE role = 'patient'
```

```
SET total_checks = COUNT(*) FROM predictions
```

```
SET responded = COUNT(DISTINCT patient_id) FROM doctor_feedback
```

```
SET pending = total_patients - responded
```

```
RETURN {total_patients, total_checks, responded, pending}
```

```
END FUNCTION
```

3.2 System UI Design

The User Interface (UI) of the HealthBridge ecosystem is designed with a focus on Clinical Clarity, Real-time Responsiveness, and Accessibility. The system employs a bifurcated dashboard strategy: one optimized for medical professionals (Physician Oversight) and another for patients (Personal Health Monitoring).

3.2.1 Doctor Dashboard

- Population Analytics: Real-time metrics on patient load and risk alerts.
- Patient Monitoring: Tabular view of Deep Learning diagnoses and risk scores.
- Feedback Portal: Interface for physicians to issue verified prescriptions.
- Search & Filter: Instant patient isolation by Institutional ID or Name.

3.2.2 Patient Dashboard

- Live Vitals: Real-time display of Temperature, BPM, SpO2, and BP.
- Diagnostic Gauge: ML-predicted status with color-coded confidence levels.
- Therapeutic Roadmap: Structured lists of Meds, Diets, and Routines.
- Analytics: Interactive charts visualizing health trends over time.

3.2.3 Llama 3 AI Chatbot

- Conversational AI: Messaging UI for natural language medical queries.
- Contextual Intelligence: Groq-Llama 3 answers grounded in MySQL history.
- Accessibility: High-contrast icons and clear typography for all users.

3.3 System Workflow

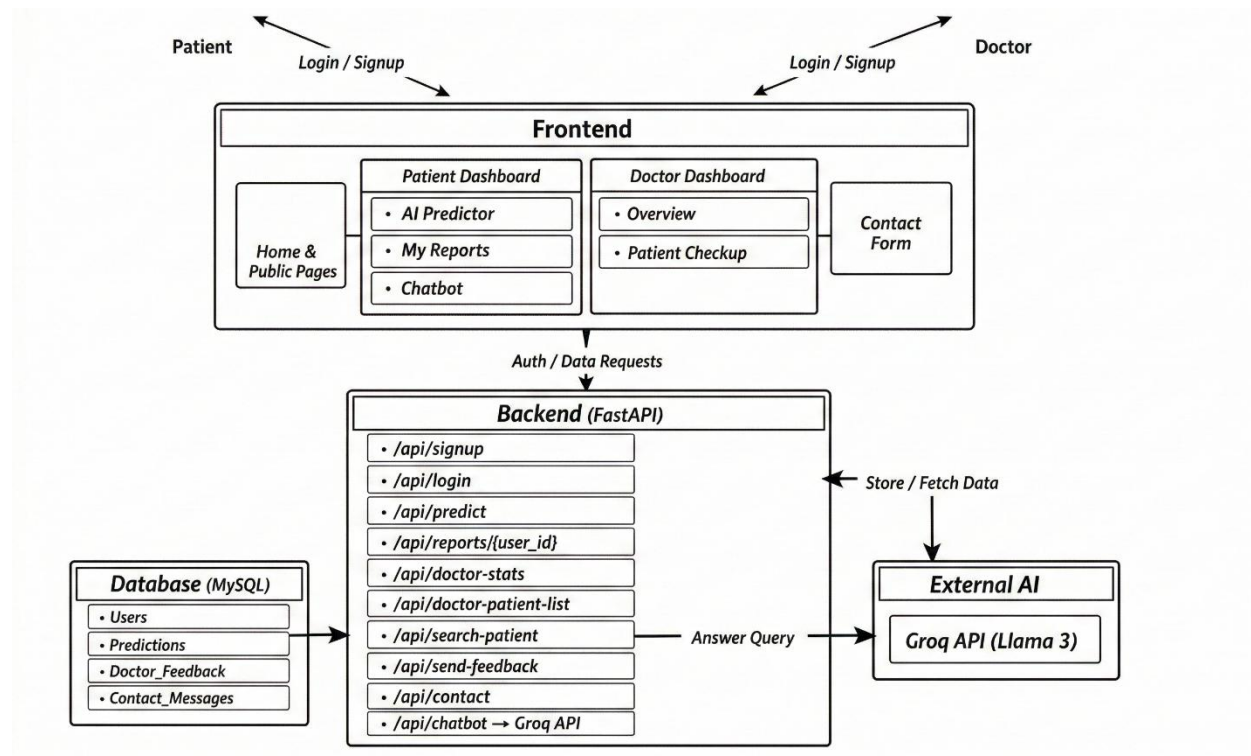


Figure 1: System Workflow Diagram

3.4 ER Diagram

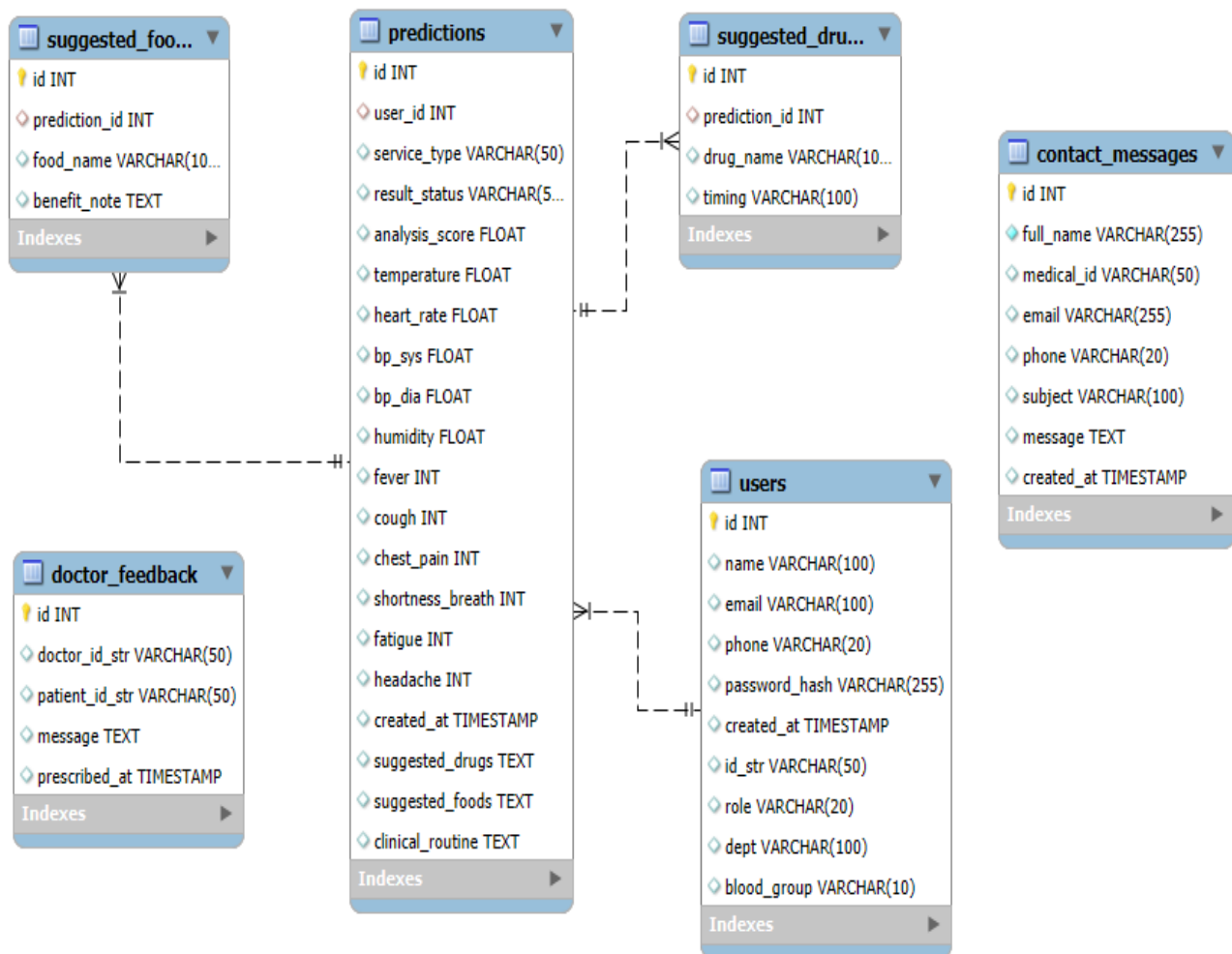


Figure 2: ER Diagram of healthbridge_db

3.5 Method

The HealthBridge methodology integrates IoT-based biometric data acquisition with a high-performance FastAPI backend to provide real-time clinical diagnostic intelligence. Physiological vitals, such as a 145.1 mmHg mean systolic BP for hypertension, are captured via ESP8266-interfaced sensors and combined with subjective symptoms quantified on a 0–10 scale for multimodal analysis. These data points are processed through a comparative machine learning architecture where Random Forest and Deep Learning models achieved a peak accuracy of 98.9%, with feature validation via the Ablation Method confirming systolic blood pressure as the most critical diagnostic variable. Finally, the system provides an interactive React-based dashboard featuring a Bioactive Food Directory to support proactive patient wellness and institutional medical support.

CHAPTER-4

RESULT AND DISCUSSION

4.1 System Interface and User Experience

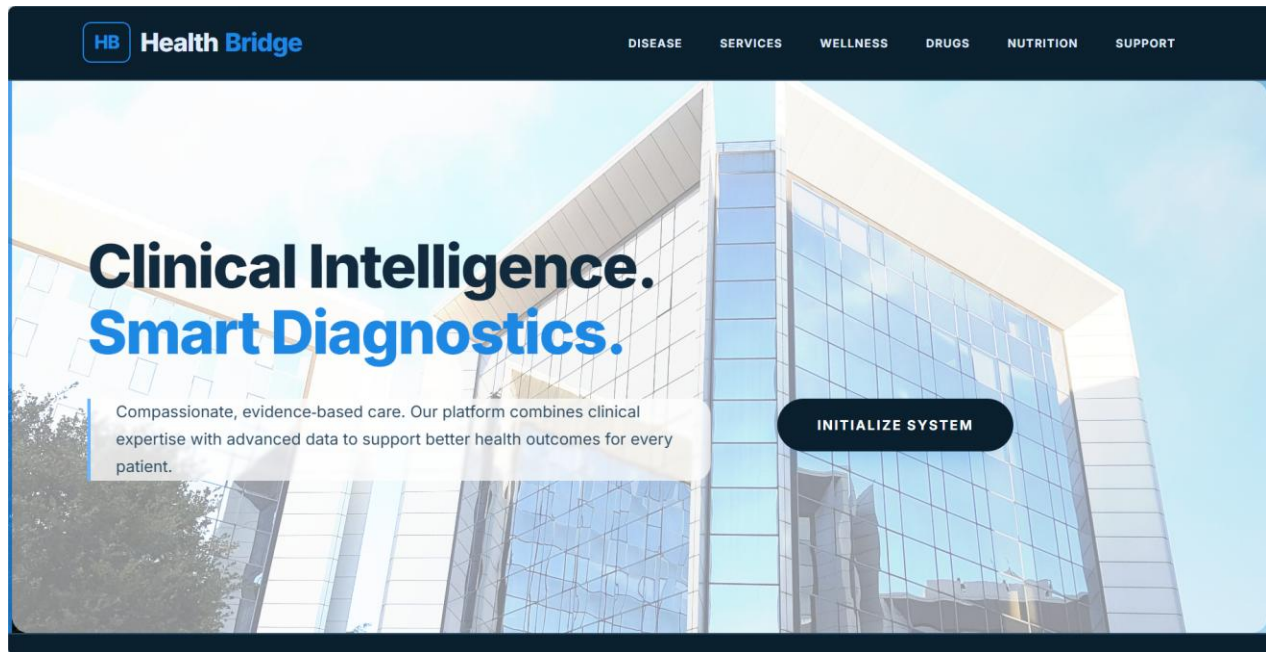
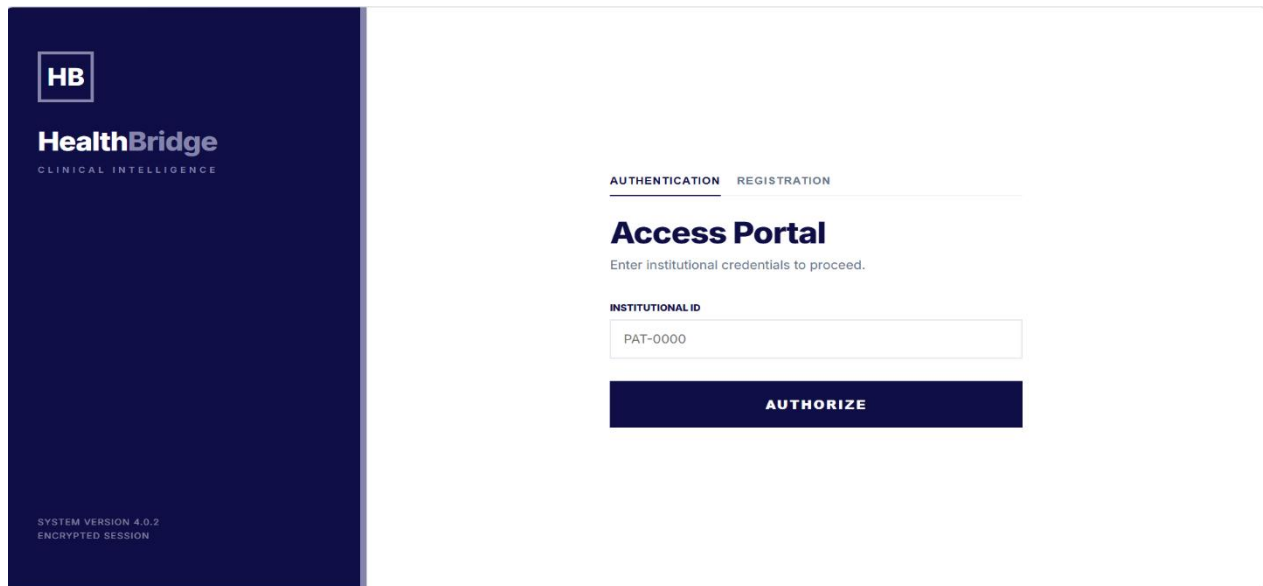


Figure 3: HealthBridge Home Interface

The HealthBridge home screen serves as a professional clinical portal featuring a Smart Diagnostics interface and a prominent Initialize System button to begin real-time monitoring. The navigation bar provides direct access to core services including disease tracking, wellness resources, drug information, and nutritional guidance.



The screenshot shows the 'Access Portal' login interface. On the left is a dark blue sidebar with the 'HB HealthBridge CLINICAL INTELLIGENCE' logo and 'SYSTEM VERSION 4.0.2 ENCRYPTED SESSION' at the bottom. The main content area has a white background with a top navigation bar containing 'AUTHENTICATION' (selected) and 'REGISTRATION'. Below this is the 'Access Portal' title and a subtitle 'Enter institutional credentials to proceed.'. A form for 'INSTITUTIONAL ID' contains the text 'PAT-0000'. At the bottom is a dark blue 'AUTHORIZE' button.

HB
HealthBridge
CLINICAL INTELLIGENCE

SYSTEM VERSION 4.0.2
ENCRYPTED SESSION

AUTHENTICATION REGISTRATION

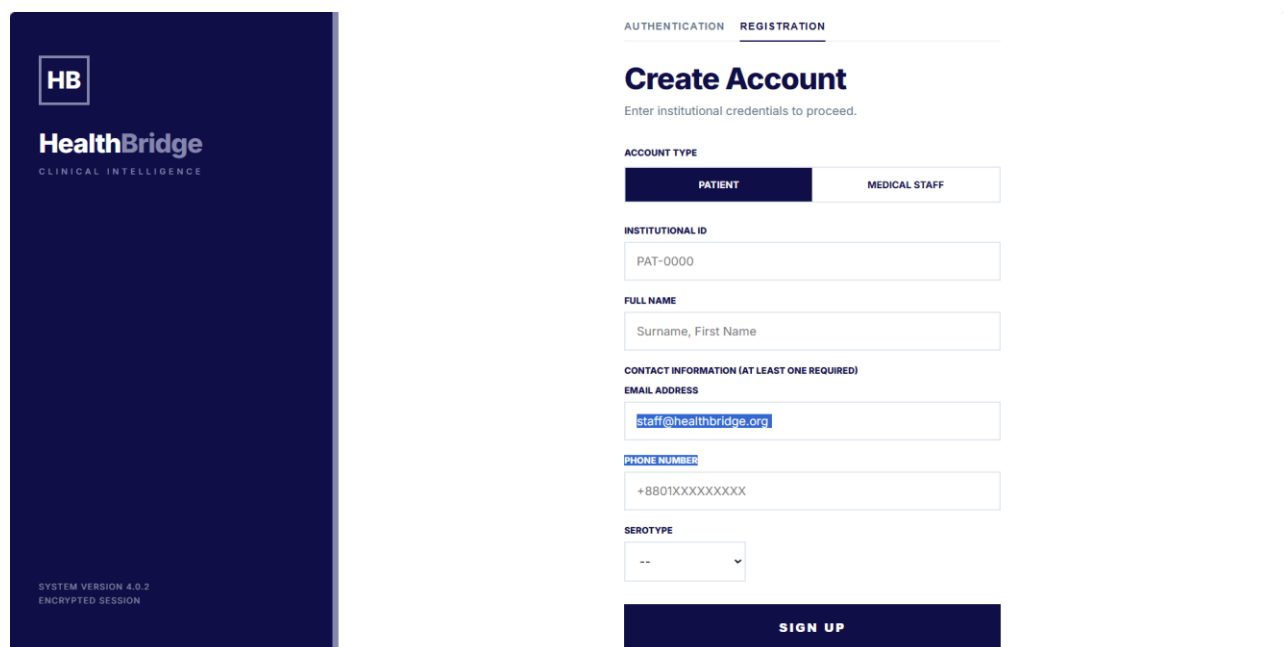
Access Portal
Enter institutional credentials to proceed.

INSTITUTIONAL ID

PAT-0000

AUTHORIZE

Figure 4: Login Interface



The screenshot shows the 'Create Account' registration interface. The sidebar is identical to Figure 4. The main content area has a white background with a top navigation bar containing 'AUTHENTICATION' and 'REGISTRATION' (selected). Below this is the 'Create Account' title and a subtitle 'Enter institutional credentials to proceed.'. The 'ACCOUNT TYPE' section has two buttons: 'PATIENT' (selected) and 'MEDICAL STAFF'. Below are form fields for 'INSTITUTIONAL ID' (PAT-0000), 'FULL NAME' (Surname, First Name), 'CONTACT INFORMATION (AT LEAST ONE REQUIRED)' with sub-fields for 'EMAIL ADDRESS' (staff@healthbridge.org) and 'PHONE NUMBER' (+8801XXXXXXXXXX), and a 'SEROTYPE' dropdown menu. At the bottom is a dark blue 'SIGN UP' button.

HB
HealthBridge
CLINICAL INTELLIGENCE

SYSTEM VERSION 4.0.2
ENCRYPTED SESSION

AUTHENTICATION REGISTRATION

Create Account
Enter institutional credentials to proceed.

ACCOUNT TYPE

PATIENT MEDICAL STAFF

INSTITUTIONAL ID

PAT-0000

FULL NAME

Surname, First Name

CONTACT INFORMATION (AT LEAST ONE REQUIRED)

EMAIL ADDRESS

staff@healthbridge.org

PHONE NUMBER

+8801XXXXXXXXXX

SEROTYPE

--

SIGN UP

Figure 5: Registration Interface

The HealthBridge Access Portal provides a secure, institutional authentication environment that supports role-based access for both patients and medical staff. The Registration interface facilitates comprehensive data onboarding by capturing unique institutional IDs, contact information, and clinical markers like blood group (serotype) within an encrypted session. Once registered, users can

securely authorize their identity through a streamlined Authentication portal, ensuring that sensitive medical dashboards and diagnostic tools remain protected and accessible only to verified individuals.

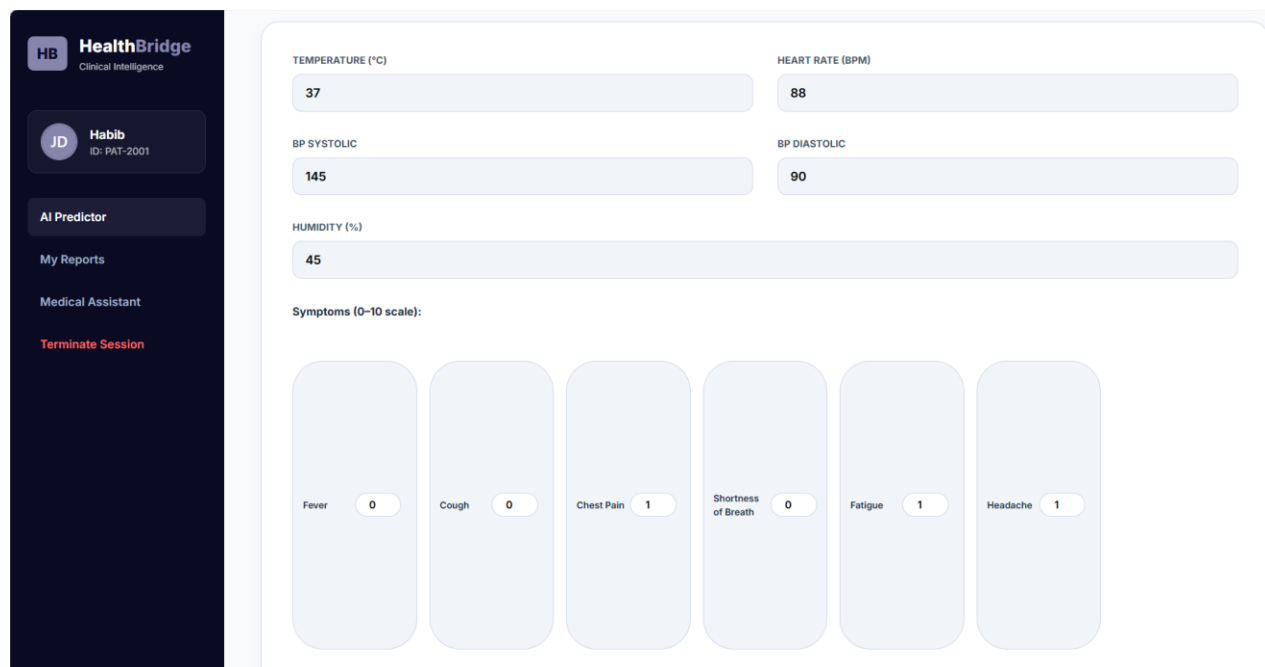


Figure 6: Real-Time Acquisition & Symptoms Logging

The AI Predictor interface provides a comprehensive health monitoring environment where real-time physiological vitals, such as Temperature, Heart Rate, and Blood Pressure, are automatically synchronized alongside a patient-reported Symptoms Log. Within this unified dashboard, users can quantify subjective symptoms on a 0–10 scale, allowing the system to combine hardware-captured data with personal health observations for more accurate analysis. The interface is supported by a secure sidebar for authenticated users, like Habib (ID: PAT-2001), ensuring seamless navigation between predictive tools, medical history, and AI-driven assistance while maintaining data privacy through session termination options.

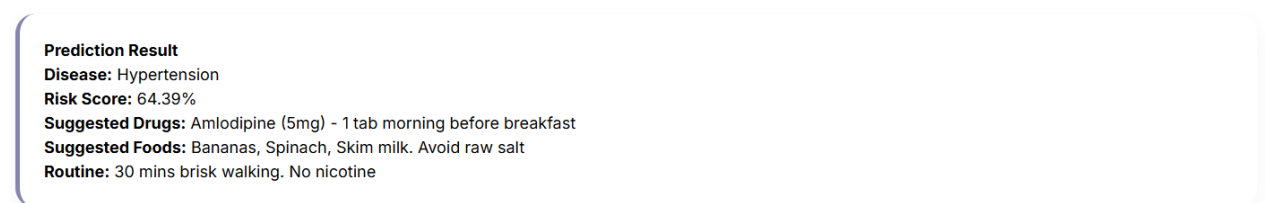


Figure 7: Clinical Prediction & Diagnostic Result

The HealthBridge Diagnostic Interface generates an automated Risk Score for identified conditions like Hypertension by analyzing real-time sensor data and patient symptoms. The results are

translated into an actionable recovery plan featuring specific Pharmacological Guidance, heart-healthy Nutrition, and targeted Lifestyle Routines. This structured output ensures patients receive immediate, data-driven clinical advice to effectively bridge the gap between monitoring and proactive health management.

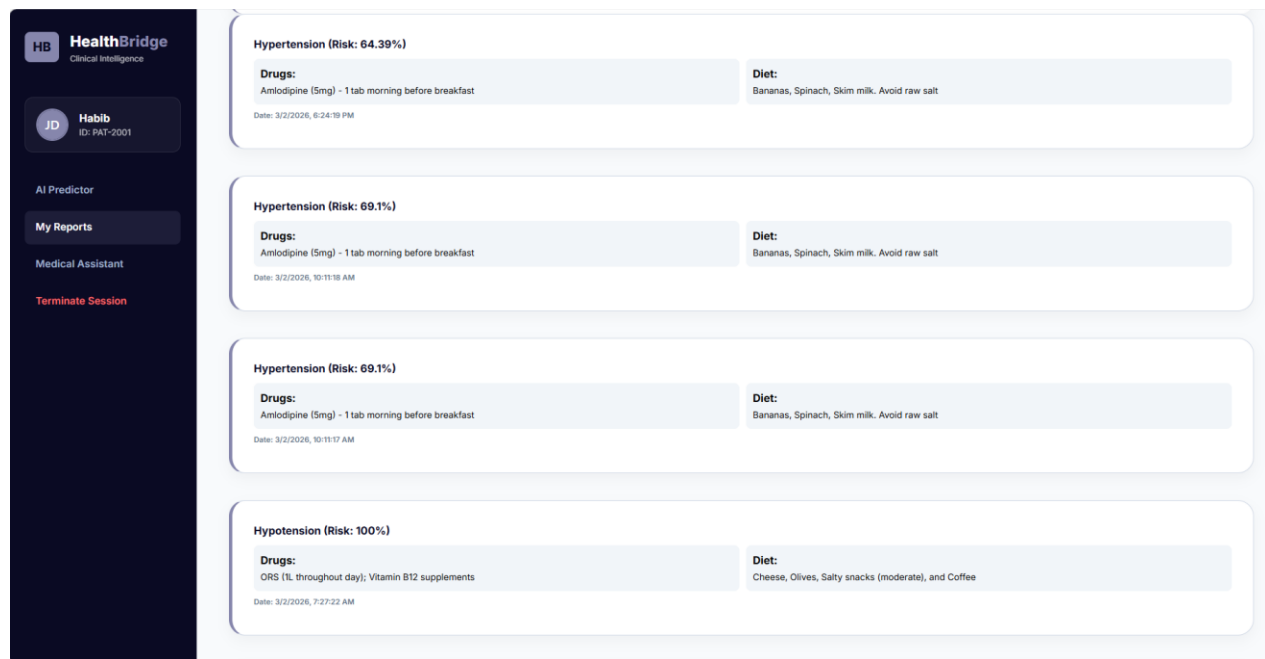


Figure 8: Historical Tracking & Longitudinal Analysis

The My Reports interface facilitates longitudinal health monitoring by archiving diagnostic sessions in chronological order, allowing users to track the progression of conditions like Hypertension and Hypotension alongside their respective risk scores. Each log provides a persistent record of system-generated Drugs and Dietary plans, enabling patients to review past clinical instructions and monitor the stability of their health markers over time. This centralized repository ensures that critical data remains accessible for both patient reference and physician review, bridging the gap between individual checkups and long-term recovery analysis.

My Reports

CLINICAL RECORDS

Diagnostic History (Latest First)

MESSAGE FROM DOCTOR

"Your records show fluctuating blood pressure (high and low) with repeated heart risk alerts. This needs close monitoring.

Continue Amlodipine 5 mg once daily as prescribed.

Do not mix BP medicines unless advised.

Monitor blood pressure twice daily and keep a record.

Use Nitroglycerin only for emergency chest pain.

Follow a strict low-salt, heart-healthy diet.

Recommended tests: ECG, lipid profile, kidney function test.

If chest pain, severe dizziness, or breathlessness occurs, seek emergency care immediately.

Follow up in 7 days."

Figure 9: Integrated Clinical Feedback & Diagnostic History

The My Reports section features a dedicated Message from Doctor portal that provides personalized clinical oversight by reviewing the patient's longitudinal diagnostic history. In this interface, physicians can issue critical medical directives such as specific medication adjustments for fluctuating blood pressure, emergency protocols for chest pain using Nitroglycerin, and a schedule for recommended diagnostic tests like ECG and lipid profiles. By integrating these direct professional insights with the patient's recorded diagnostic logs, the system ensures a continuous feedback loop between automated AI predictions and expert human intervention for enhanced patient safety.

MEDICAL ASSISTANT

Chat with AI for health guidance

Hello! I'm your medical assistant. How can I help you today?

"Your records show fluctuating blood pressure (high and low) with repeated heart risk alerts. This needs close monitoring. Continue Amlodipine 5 mg once daily as prescribed. Do not mix BP medicines unless advised. Monitor blood pressure twice daily and keep a record. Use Nitroglycerin only for emergency chest pain. Follow a strict low-salt, heart-healthy diet. Recommended tests: ECG, lipid profile, kidney function test. If chest pain, severe dizziness, or breathlessness occurs, seek emergency care immediately. Follow up in 7 days." Doctor says this, what I should do?

Based on the doctor's instructions, you should: 1. Continue taking Amlodipine 5mg once daily as prescribed. 2. Monitor your blood pressure twice daily and keep a record. 3. Follow a strict low-salt, heart-healthy diet. 4. Use Nitroglycerin only in case of emergency chest pain. 5. Undergo recommended tests: ECG, lipid profile, and kidney function test. 6. Seek emergency care immediately if you experience chest pain, severe dizziness, or breathlessness. 7. Schedule a follow-up appointment in 7 days. Remember not to mix blood pressure medicines unless advised by the doctor.

Type your question...

Send

Figure 10: AI Medical Assistant & Contextual Conversational Guidance

The Medical Assistant serves as an interactive conversational interface where patients can engage with an AI Chatbot for immediate health guidance based on their diagnostic history. By utilizing large language model inference through the Groq Cloud, the assistant can interpret complex medical directives such as physician messages regarding fluctuating blood pressure and emergency chest pain protocols and translate them into clear, prioritized action steps for the user. This feature ensures that clinical instructions remain accessible and easy to follow, providing a continuous layer of automated support that complements the system's real-time monitoring and historical reporting capabilities.

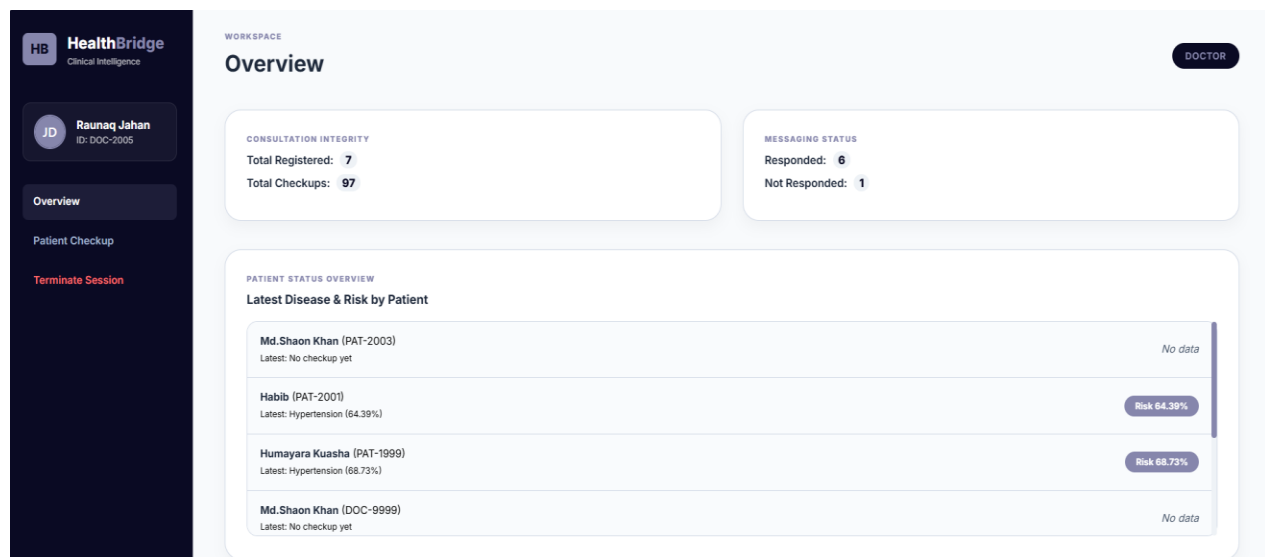


Figure 11: Doctor Workspace & Clinical Oversight Dashboard

The Doctor Workspace provides a centralized overview of clinical integrity, allowing physicians like Raunaq Jahan (ID: DOC-2005) to monitor patient checkup statistics and manage consultation responses in real-time. This interface includes a Patient Status Overview that tracks the latest disease predictions and risk scores across a registered user base, such as identifying a 64.39% risk of Hypertension for specific patients. By aggregating these metrics into a high-level summary, the dashboard enables medical staff to prioritize critical cases and maintain efficient communication across the healthcare platform.

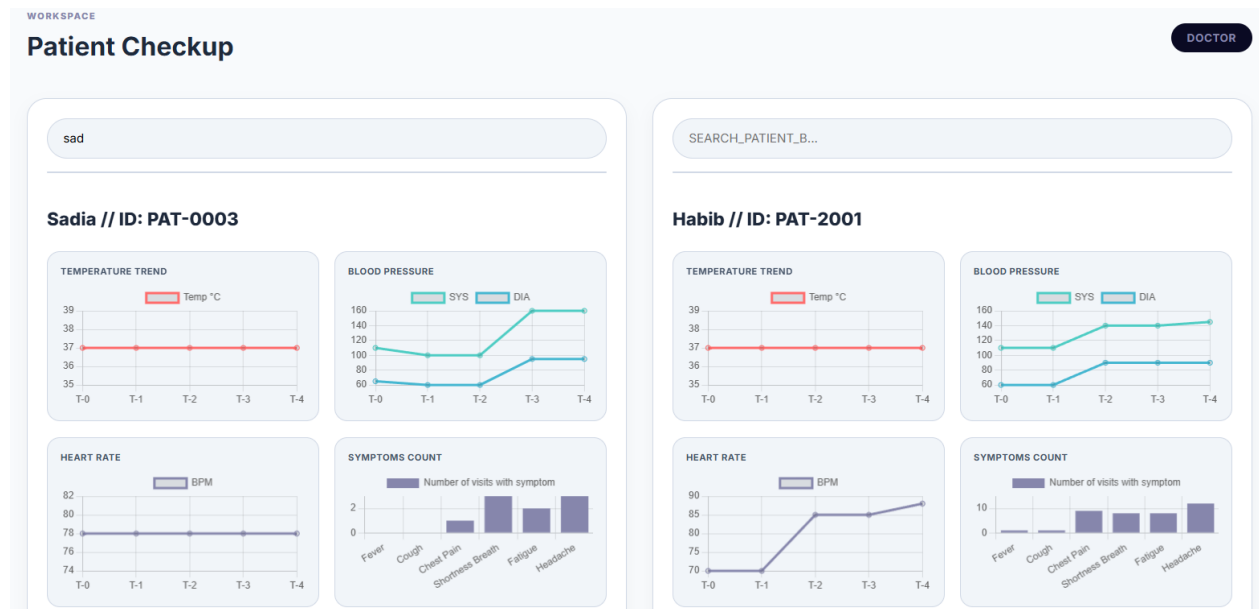


Figure 12: Comprehensive Patient Checkup & Trend Analysis

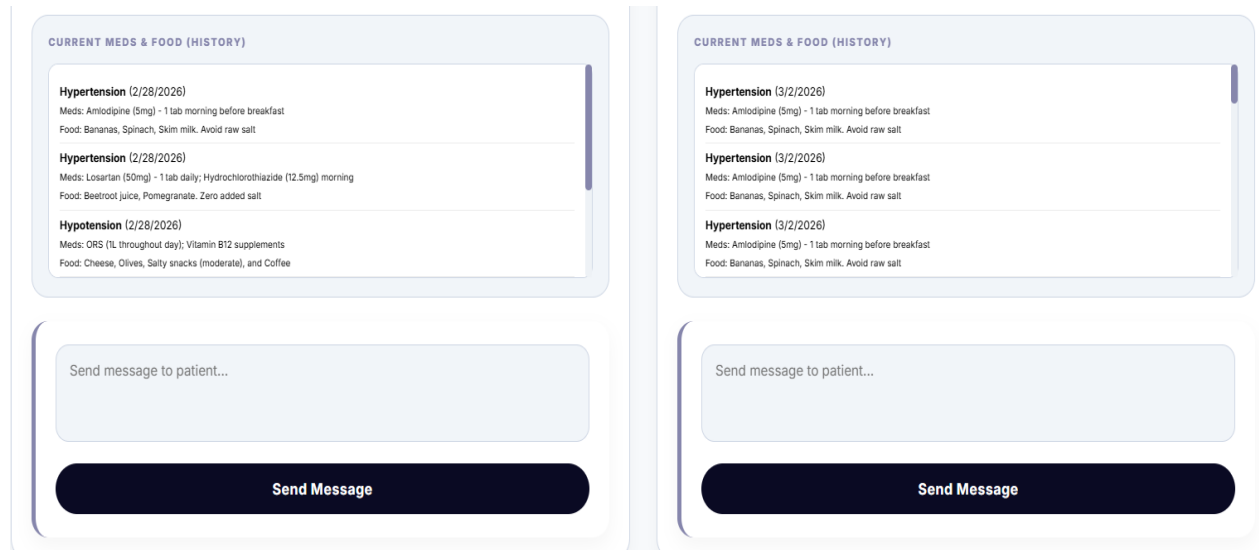


Figure 13: Clinical Messaging Interface & Historical Medication Review

The Doctor Workspace enables comprehensive clinical oversight by combining real-time trend analysis with a direct intervention portal. Through the Patient Checkup dashboard, medical staff can monitor longitudinal vitals such as Temperature, Blood Pressure, and Heart Rate across five time intervals while using Symptoms Count charts to identify emerging health patterns. This objective data is paired with a Physician Consultation interface where doctors review historical medications and dietary logs to issue immediate, personalized medical guidance via a dedicated

messaging system. By integrating visual analytics with direct communication, the system ensures that automated monitoring is consistently supported by expert human intervention.

Recent Advances in Oncology

Cancer research is moving rapidly. Here are three areas where patients are already seeing benefits.

CAR-T cell therapy

In this approach, a patient's own T cells are engineered to recognize and attack cancer cells. It has produced remarkable responses in certain leukemias and lymphomas, and clinical trials are now testing it against solid tumors like glioblastoma and sarcoma.

Liquid biopsies

A simple blood test can now detect tiny fragments of tumor DNA. This allows doctors to monitor how well treatment is working, spot recurrence earlier, and even identify which targeted therapies might be effective—all without an invasive tissue biopsy.

AI-assisted pathology

Machine learning algorithms are being trained to examine biopsy slides. In studies, they match or even exceed the accuracy of expert pathologists in detecting breast cancer metastases and lung cancer subtypes, potentially speeding up diagnosis and reducing errors.

Clinical trial opportunities: Many of these new treatments are available only through research studies. Patients should ask their oncologist about trials that might be appropriate for their specific cancer type and stage.



ADVANCED CLINICAL REPORT

Oncology: Pathophysiology & Molecular Therapeutics

Figure 14: Advanced Clinical Research & Oncology Insights

In the Disease section of the main navigation bar, the platform catalogues some of the most common and life-threatening conditions to provide users with in-depth research details and classifications. This specialized portal, such as the Advanced Clinical Report on Oncology, offers comprehensive insights into pathophysiology and molecular therapeutics while highlighting recent medical breakthroughs like CAR-T cell therapy, Liquid biopsies, and AI-assisted pathology. By integrating these research-heavy resources directly into the platform, the project serves not only as a monitoring tool but also as a vital educational hub for understanding complex disease mechanisms and emerging treatment opportunities.

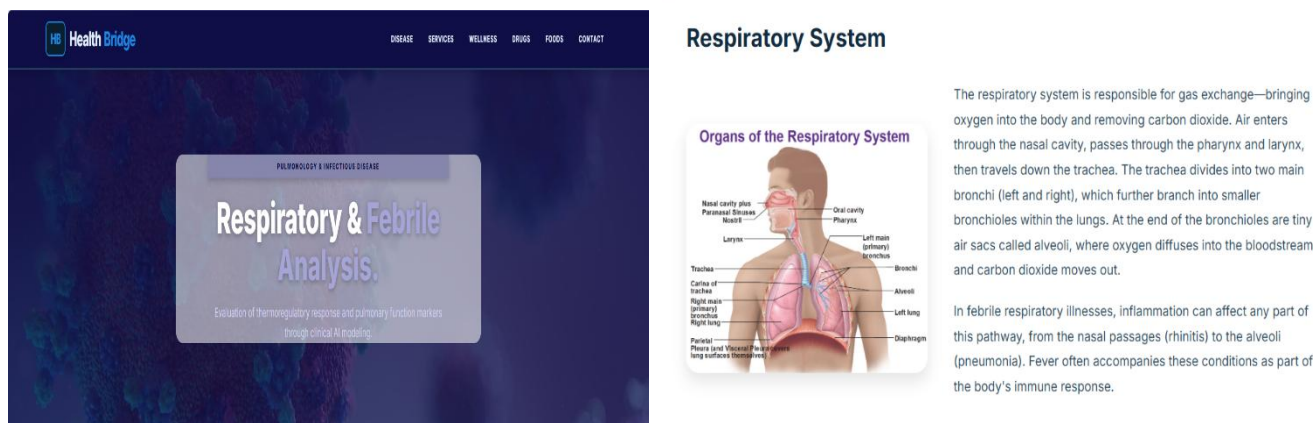


Figure 15: Clinical Services and Specialized Disease Analysis

The Services section of the HealthBridge platform provides an expansive library of critical health conditions, including Heart Attack, Respiratory Fever, Hypertension, and Hypotension. These modules are designed to bridge the gap between automated monitoring and clinical education through detailed physiological analysis.

Respiratory and Febrile Analysis Service Interface The interface provides a specialized portal for Pulmonology and Infectious Disease, utilizing clinical AI modeling to evaluate thermoregulatory responses and pulmonary function markers. It offers a high-level overview of how fever and respiratory illnesses impact the body's immune response.

Anatomical Mapping and Respiratory Pathophysiology This component provides a detailed anatomical breakdown of the Respiratory System, tracing the path of gas exchange from the nasal cavity to the alveoli. The system analyzes how inflammation affects different parts of this pathway such as rhinitis in nasal passages or pneumonia in the alveoli enabling users to understand the physiological basis of the symptoms tracked by the hardware sensors.

Restorative Glymphatic Function

Sleep is a metabolic necessity. During non-REM stages, the brain's glymphatic system clears beta-amyloid proteins. Disrupting this rhythm leads to metabolic dysregulation and reduced cellular repair. Adults need 7–9 hours per night; consistent sleep timing reinforces the circadian clock and optimizes hormone release.

PHASE
1

Maintain strict circadian consistency—same bedtime and wake time daily.

PHASE
2

Eliminate short-wave (blue) light exposure at least one hour before sleep.

PHASE
3

Optimize thermoregulation (cool room, 18–20°C) to facilitate deep REM.



Figure 16: Personalized Wellness & Preventive Science

The Wellness module provides evidence-based clinical strategies for maintaining physiological equilibrium, focusing on key indicators such as Glymphatic Function, Heart Rate Variability (HRV), and Circadian Alignment. It offers actionable phases for restorative health, including strict circadian consistency and optimized thermoregulation to facilitate deep REM sleep and metabolic repair. By integrating wearable biofeedback and personalized nutrition guidelines, the system empowers users to manage chronic stress and improve cardiovascular longevity through consistent, data-driven lifestyle adjustments.

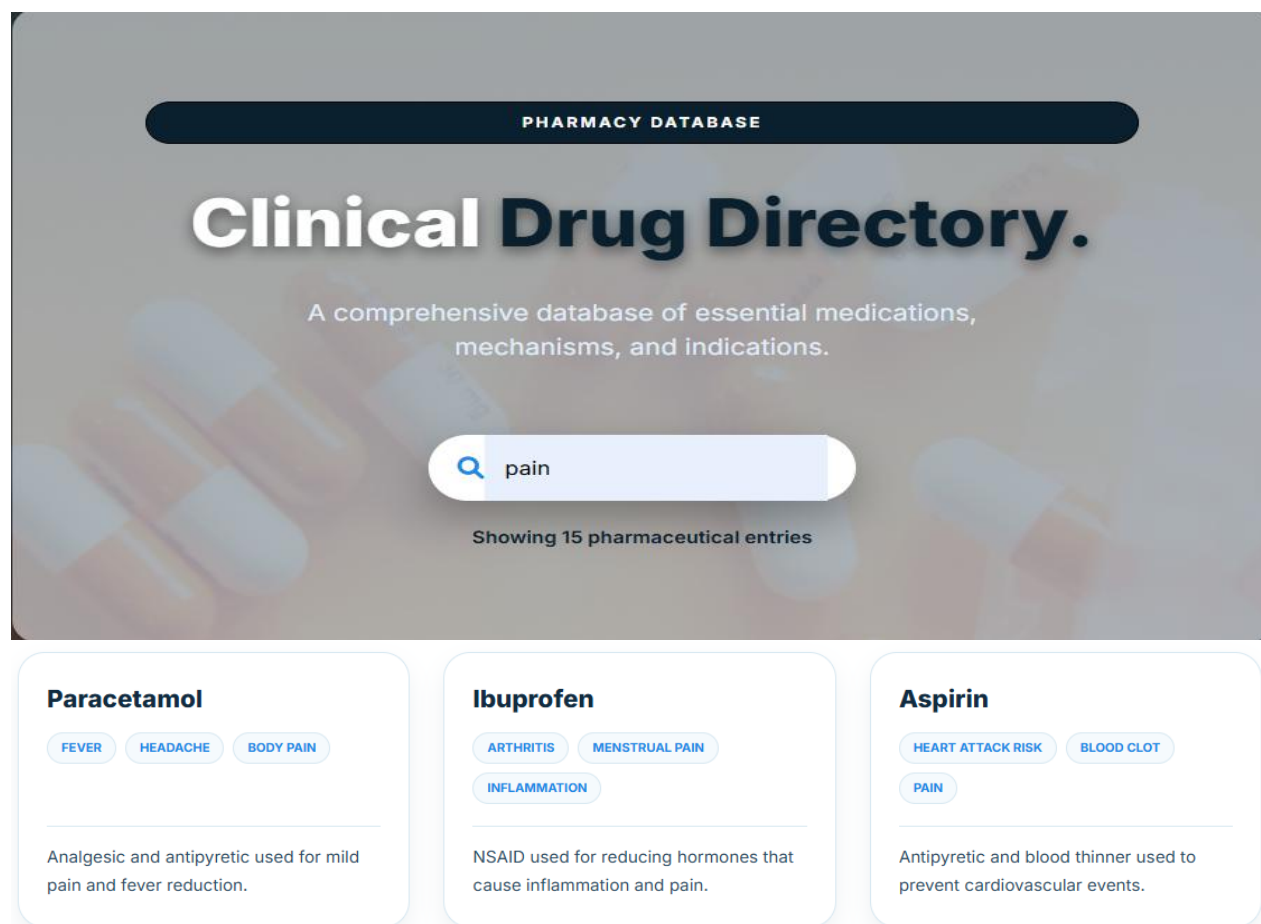


Figure 17:Pharmaceutical Directory & Medication Management

The Drugs section features a comprehensive Clinical Drug Directory that provides patients and healthcare providers with a searchable database of essential medications, their mechanisms of action, and primary indications. Categorized into major classes such as Antihypertensives (e.g., Amlodipine), Analgesics (e.g., Paracetamol), and Antibiotics (e.g., Amoxicillin), each entry is designed to improve medication adherence and safety through clear educational briefs. By integrating standardized pharmacological references with safe medication practices including guidance on drug interactions and side-effect reporting the portal ensures that users have immediate access to high-fidelity medical information that supports their personalized treatment plans.



Figure 18: Nutritional Therapeutics & Bioactive Compound Database

The Foods section, titled Nature's Pharmacy, serves as a data-driven repository of bioactive compounds designed to prevent and manage chronic clinical conditions through targeted nutrition. This interface categorizes whole foods by their Clinical Target, such as utilizing Allicin from Garlic for hypertension, Curcumin from Turmeric for chronic inflammation, and Iron-rich Leafy Spinach for anemia prevention. By integrating evidence-based principles like Bioavailability Considerations for example, pairing curcumin with healthy fats to enhance absorption the system provides patients with actionable dietary strategies that complement their pharmacological treatments and support long-term physiological stability.



Figure 19: Clinical Support & Multi-Channel Communication

The Contact section provides a centralized hub for Clinical Support, ensuring that both patients and medical professionals have direct access to administrative and technical assistance. The interface features a structured Inquiry Form that allows users to categorize their requests ranging from Technical Support and Billing to specific Clinical Questions facilitating efficient routing to the appropriate department. To ensure continuous care, the portal lists 24/7 direct contact channels, including a dedicated main switchboard and specialized email support for technical issues like portal navigation or AI module troubleshooting. Additionally, the section incorporates a critical Emergency Notice and a Frequently Asked Questions (FAQ) repository, providing immediate guidance on accessing clinical records and managing life-threatening situations.

4.2 Exploratory Data Analysis

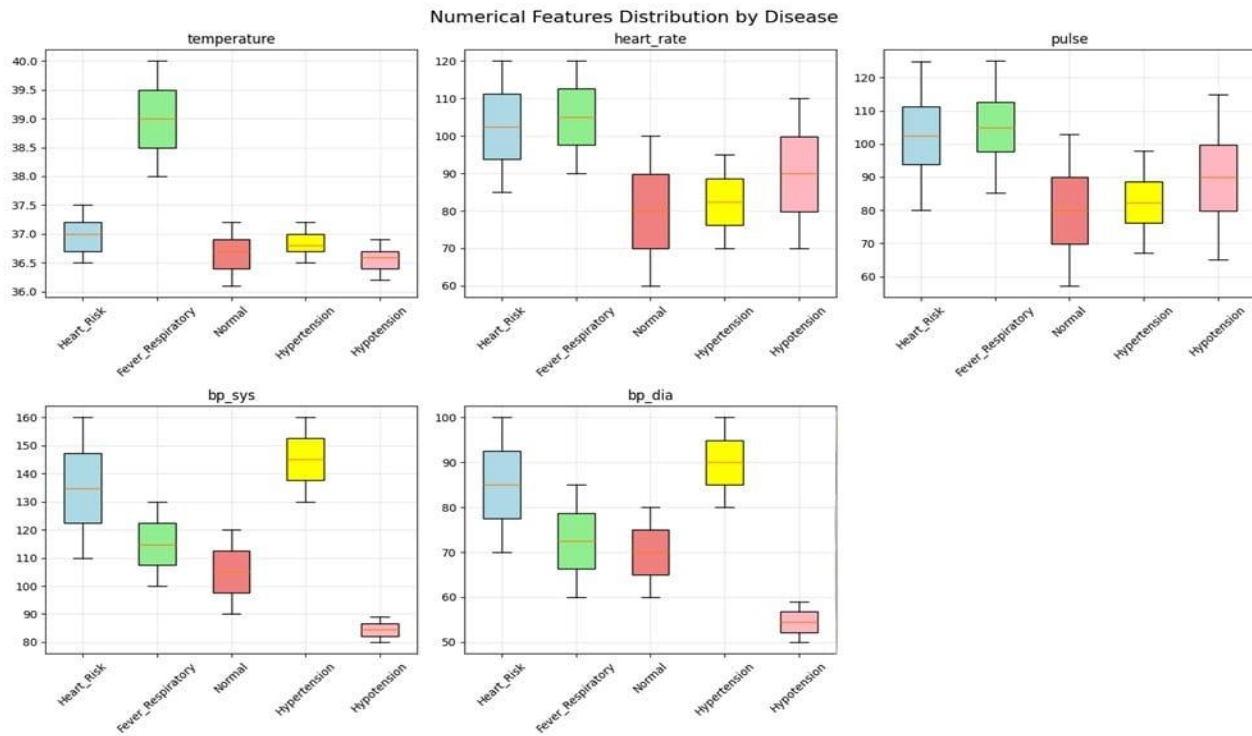


Figure 20: Numerical Features Distribution By Disease

This visualization displays how vital signs like temperature, heart rate, and blood pressure vary across different medical conditions.

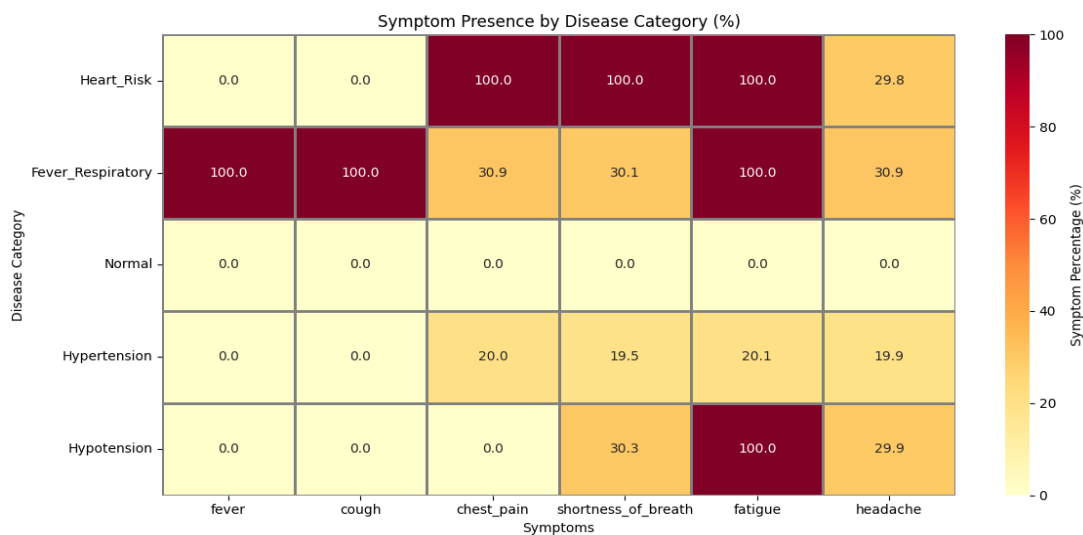


Figure 21: Symptoms Presence by Disease Category

This heatmap illustrates the percentage of symptom presence for each disease category, highlighting distinct patterns like Fever Respiratory being uniquely defined by fever and cough. Meanwhile, fatigue appears as a universal symptom across all conditions except normal, which shows a complete absence of the listed symptoms.

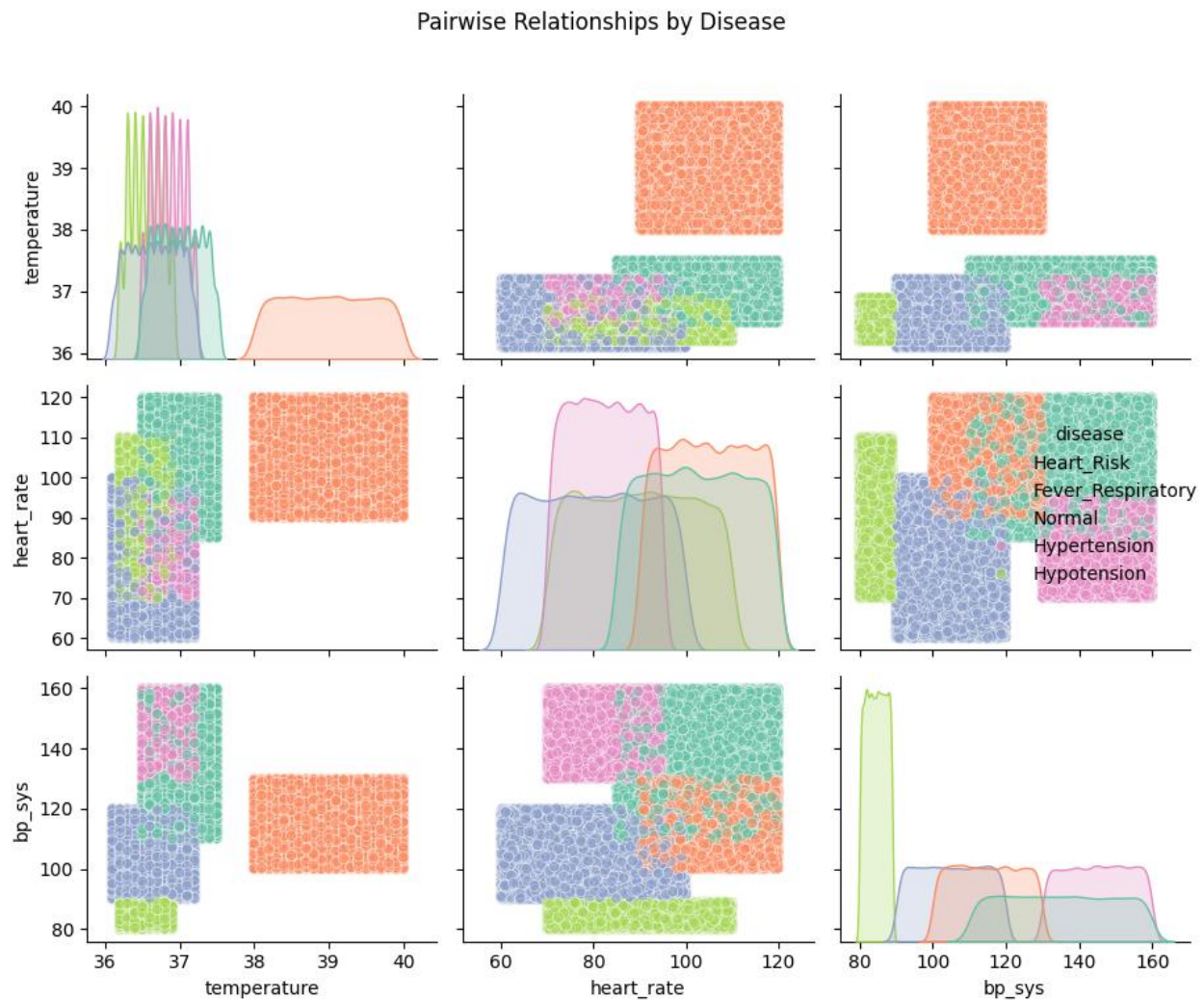


Figure 22: Pairwise Relationships of Vital Signs by Disease Category

This pair plot visualizes how vital signs like temperature, heart rate, and blood pressure cluster differently for each condition, such as fever respiratory showing a distinct separation due to high temperature. These clear distributions across the scatter plots and density curves indicate that the variables are effective features for distinguishing between categories like heart risk and Normal for the classification task.

4.3 Model Performance and Evaluation

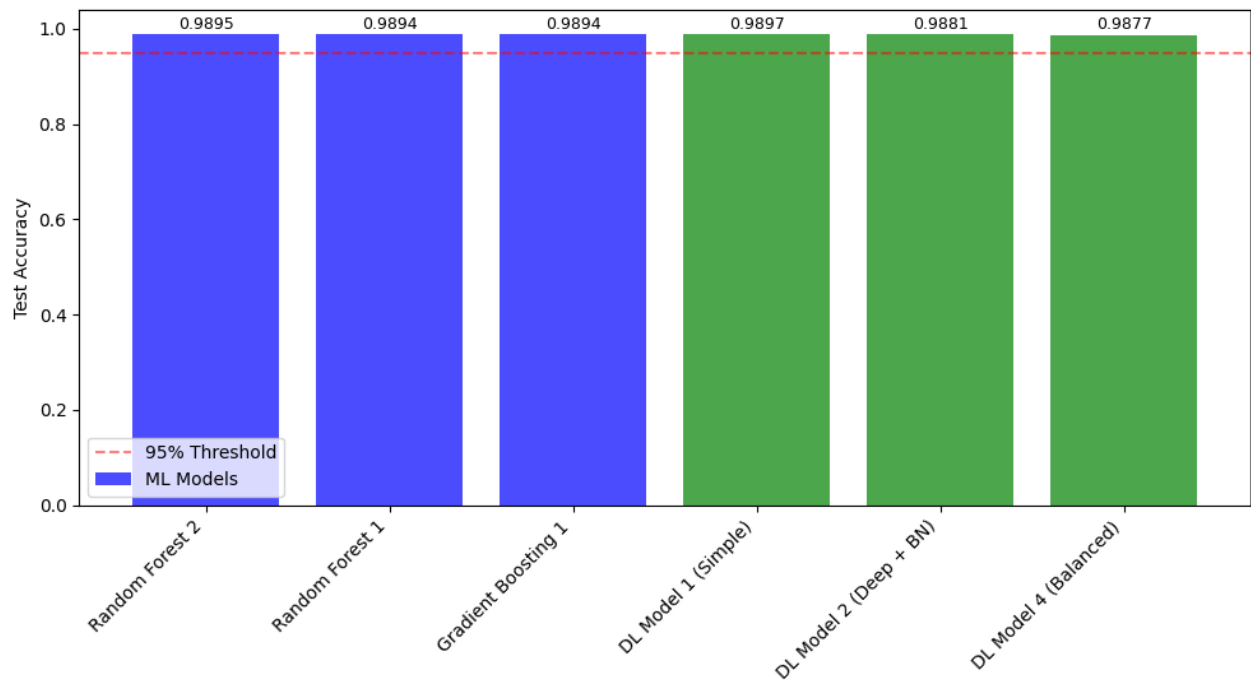


Figure 23: Best Performing Models: ML vs DL Comparison

This bar chart compares the test accuracy of top Machine Learning (ML) and Deep Learning (DL) models for the IoT-based disease prediction system. All models, including Random Forest and Simple DL, achieved exceptionally high accuracy scores exceeding 98.9%, significantly surpassing the established 95% performance threshold. This demonstrates that feature selection incorporating both vital signs and symptom presence provides a highly robust foundation for accurate disease classification.

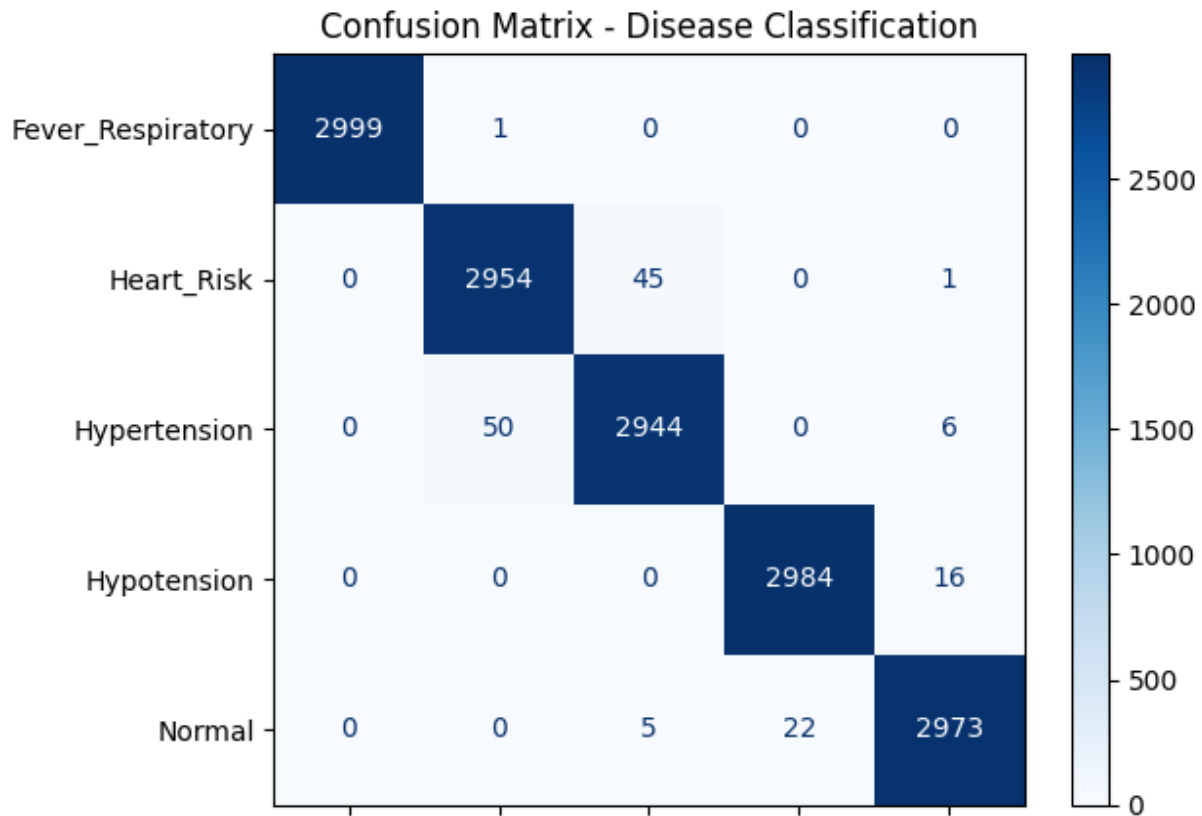


Figure 24: Confusion Matrix - Disease Classification

This confusion matrix demonstrates the model's high classification reliability, showing near-perfect diagonal alignment where almost all cases, such as Fever Respiratory (2,999/3,000), were correctly predicted. The minimal misclassifications, such as 45 Heart Risk cases being identified as Hypertension, reflect the slight statistical overlap in vital signs between these cardiovascular categories. Overall, the matrix validates the exceptional 98.7%+ test accuracy across the machine learning and deep learning models.

CHAPTER-5

CONCLUSION

5.1 Future Work

The HealthBridge platform establishes a foundational framework for AI-driven clinical intelligence, yet future iterations will focus on scaling these capabilities through the integration of high-fidelity, decentralized datasets to broaden the diagnostic repertoire for complex diseases. A primary objective is to transition toward a mobile-first, HIPAA-compliant architecture that supports seamless synchronization with smartphones and smartwatches, enabling continuous, 24/7 monitoring of critical vitals like SpO₂, Heart Rate Variability (HRV), and Blood Pressure directly from wearable devices.

Future development will prioritize a multimodal diagnostic environment by implementing Convolutional Neural Networks (CNNs) for the automated analysis of X-rays and MRI scans, alongside Recurrent Neural Networks (RNNs) for processing real-time ECG waveforms to identify arrhythmias. Beyond imaging, we plan to incorporate AI-driven blood report analysis to cross-reference lipid profiles and glucose levels against existing vitals, while implementing edge computing on the hardware level for local anomaly detection during network latency. By expanding the Message from Doctor portal into a full-scale telemedicine suite and utilizing federated learning to refine therapeutic suggestions, the system will evolve into a proactive, life-long health management ecosystem that supports both automated AI pathology and expert human oversight.

5.2 Conclusion

The HealthBridge project successfully demonstrates the integration of IoT-based hardware with advanced clinical intelligence to create a seamless, real-time health monitoring ecosystem. By bridging the gap between raw biometric data and actionable medical insights, the system empowers patients with AI-driven risk scoring, personalized nutritional guidance, and direct access to pharmacological data. Furthermore, the dedicated physician workspace ensures that automated predictions are consistently supported by expert human oversight, facilitating a collaborative approach to chronic disease management. Ultimately, this research provides a scalable and secure framework for the future of digital healthcare, moving the industry closer to a proactive, data-driven model that prioritizes early intervention and long-term physiological stability.

References

- [1] S. M. Riazul Islam, D. Kwak, M. H. Kabir, M. Hossain, and K. S. Kwak, The Internet of Things for Health Care: A Comprehensive Survey, *IEEE Access*, vol. 3, pp. 678-708, 2015, doi: 10.1109/ACCESS.2015.2437951.
- [2] A. M. Rahmani et al., Exploiting smart e-Health gateways at the edge of healthcare Internet-of-Things: A fog computing approach, *Future Generation Computer Systems*, vol. 78, pp. 641-658, 2018.
- [3] M. S. Hossain and G. Muhammad, Cloud-assisted Industrial Internet of Things (IIoT) Enabled framework for health monitoring, *Computer Networks*, vol. 101, pp. 192-202, 2016.
- [4] IEEE Computer Society, IEEE Standard for Information Technology Telecommunications and Information Exchange between Systems Local and Metropolitan Area Networks Specific Requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications, *IEEE Std 802.11-2020*, 2021.
- [5] A. H. Chen, S. Y. Huang, P. S. Hong, C. H. Cheng, and E. J. Lin, Disease Prediction by Machine Learning Over Big Data From Healthcare Communities, *IEEE Access*, vol. 5, pp. 8869-8879, 2017.
- [6] E. J. Topol, High-performance medicine: the convergence of human and artificial intelligence, *Nature Medicine*, vol. 25, no. 1, pp. 44-56, 2019.
- [7] A. Y. Hannun et al., Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network, *Nature Medicine*, vol. 25, no. 1, pp. 65-69, 2019.
- [8] C. M. Bishop, *Pattern Recognition and Machine Learning*, New York: Springer, 2006.
- [9] F. Chollet, *Deep Learning with Python*, Shelter Island, NY: Manning Publications, 2017.
- [10] S. Ramírez, FastAPI: High performance, easy to learn, fast to code, ready for production, FastAPI Official Documentation, 2024. [Online]. Available: <https://fastapi.tiangolo.com/>.
- [11] Meta Platforms, Inc., React – A JavaScript library for building user interfaces, React Official Documentation, 2024. [Online]. Available: <https://react.dev/>.
- [12] L. Richardson and S. Ruby, *RESTful Web Services*, Sebastopol, CA: O'Reilly Media, 2007.

- [13] National Center for Biotechnology Information (NCBI), PubChem Compound Database, U.S. National Library of Medicine, 2024. [Online]. Available: <https://pubchem.ncbi.nlm.nih.gov/>
- [14] U.S. Department of Health and Human Services, The Health Insurance Portability and Accountability Act of 1996 (HIPAA) Privacy and Security Rules, 2024. [Online]. Available: <https://www.hhs.gov/hipaa/>.
- [15] J. C. Dunn et al., Wearable sensors enable personalized predictions of clinical laboratory measurements, *Nature Medicine*, vol. 27, no. 6, pp. 1105-1112, 2021.
- [16] N. Rieke et al., The future of digital health with Federated Learning, *npj Digital Medicine*, vol. 3, no. 119, 2020.
- [17] Institute of Medicine (US) Committee to Review Dietary Reference Intakes, *Dietary Reference Intakes for Calcium and Vitamin D*, Washington (DC): National Academies Press (US), 2011.